

OMAR NOURI

Senior Data Scientist | AI Engineer | Backend Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist and AI Engineer with 4+ years of experience building NLP systems, conversational AI agents, and scalable data pipelines. Hands-on expertise with Python, AWS (Lambda, S3), MySQL, and machine learning frameworks including PyTorch, TensorFlow, and Hugging Face Transformers. Built RASA-based dialogue systems pre-LLM era with deep understanding of backend architecture and ML model deployment. Proven track record delivering production systems serving 10,000+ users.

PROFESSIONAL EXPERIENCE

Data Scientist / Machine Learning Engineer | DXC Technology | Rabat, Morocco (Hybrid) *Jan 2024 – Present*

- Led identification and deployment of LLM and generative AI use cases in insurance, improving decision-making and customer interactions on AWS infrastructure
- Improved RAG system performance by refining reranking algorithms, implementing hybrid search, and conducting prompt engineering — achieving 10-second response time reduction and 20% accuracy boost
- Designed and implemented agentic AI workflows and developed POCs aligned with current agentic architecture trends
- Built image and graph extraction features using LLMs; designed Excel parsing and embedding pipeline
- Conducted ongoing research on LLM and GPT advancements, integrating state-of-the-art techniques into production insurance solutions

Data Scientist | Digital Place | Tangier, Morocco (On-site) *Oct 2021 – Dec 2023*

- Won fintech challenge by building a Darija-based conversational banking assistant using the RASA framework with custom NLP models
- Built and deployed NLP models achieving 95% accuracy for French and Arabic transcription, serving 10,000+ banking app users
- Designed and maintained Airflow-based data pipelines processing 500,000+ Moroccan job postings with automated ETL workflows
- Applied transformer models (BERT, GPT, LLaMA) for custom document processing and language understanding solutions
- Represented company at GITEX; trained clients on deployed solutions and built automated PowerBI dashboards

Data Engineer Intern | Digital Place | Tangier, Morocco (On-site) *Mar 2021 – Sep 2021*

- Built first version of data scraping and analysis pipelines for 100,000+ Moroccan job board listings
- Designed and optimized MySQL database schema for storing job postings and extracted insights at scale
- Collaborated with cross-functional teams to translate business requirements into data-driven pipeline outputs

KEY PROJECTS

LLM-Based RAG Insurance Assistant — DXC Technology | 2024–2025

Built production RAG system for insurance document Q&A using LLMs on AWS. Optimized retrieval pipeline with hybrid search and reranking for accuracy and latency.

Darija Smart Banking Assistant — Digital Place | 2023

Fine-tuned Meta's Wav2Vec model on Darija dialect data; implemented action-based banking assistant using RASA framework. Won Fintech Catalyst Project challenge.

Prodige (Mouakaba) — Job Market Intelligence Platform | 2022

End-to-end platform scraping, classifying, and extracting skills from 500K+ job postings via automated Airflow pipelines; visualized insights via PowerBI dashboards.

AI-Enhanced Startup Search Engine | 2022

Fine-tuned attention-based RNN model on startup descriptions to generate keywords and recommendations, improving search result relevance.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL

Cloud & Infrastructure: AWS (Lambda, S3), GCP, Docker, Apache Airflow, Git, GitHub, Bitbucket CI/CD

Databases: MySQL, PostgreSQL, MongoDB, SQLite, Neo4j

ML / DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, spaCy, NLTK, LangChain, RASA

NLP: Text Classification, NER, Sentiment Analysis, Summarization, Translation, Seq2Seq, Language Modeling

ML Concepts: Regression, Classification, Clustering, Time Series, Transfer Learning, Reinforcement Learning, Recommender Systems

Neural Networks: RNN, LSTM, GRU, CNN, Transformers, Attention Mechanisms, Wav2Vec

ETL & BI: KNIME, Pentaho, Talend, PowerBI, Tableau, QlikView

Data Processing: Data Cleaning, Transformation, Integration, Pipeline Design, Embedding

EDUCATION

State Engineer Diploma (MSc) — Computer Engineering, Information Systems | University of Abdelmalek Essaadi — ENSA | Tangier, Morocco *Sep 2021*

Baccalaureate — Mathematical Sciences (Option B) | AL Amana Institute | Tangier, Morocco *Jun 2015*

LANGUAGES

Arabic: Native | **French:** Professional (DALF C1) | **English:** Professional